

Multi-Format World Clock Dashboard

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Some airports and hotels have clocks showing different times in different countries for the convenience of people from those time zones. Sometimes the clocks are analogue and sometimes digital. The time at various places, of course, varies with the sun's position with respect to earth's longitude and latitude. Hence, to calculate time at a place, an international date line (IDL) near Wellington in New Zealand has been considered as the central location. The countries east of IDL are in time zones that are yet to be reached at IDL, while countries west of IDL are in time zones that have already been reached at IDL. That is, the countries in West lag in time with respect to the countries in East, and the time lag widens with the distance between them.

Described here is a clock program that can display times in different countries in analogue as well as digital formats. Through menu selection in the program, the interface shows the time corresponding to the selected country. The program (adwd.htm) has been coded in JavaScript, which takes care of updating the time dynamically.

The program has two parts, one of which shows the digital dashboard while the other displays the clock in analogue form. The program, when launched, displays the interface as shown in Screenshot 1.

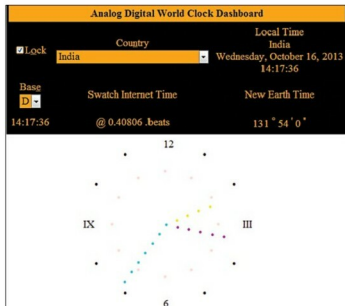
Digital clock

The digital dashboard has six functions: Lock, Country, Local Time, Base Time, Swatch Internet Time and New Earth Time. The analogue clock is displayed below this dashboard.

Lock. This function locks and unlocks the movement of analogue clock. By default, it is locked. When unlocked, the analogue clock moves

and tracks the mouse cursor. It can be accessed by pressing Alt+O from the keyboard.

Country. This displays a box with a list of countries that this program can display. The offset time is set by taking the Indian standard time (IST)



Screenshot 1: Program output of multi-format world clock dashboard

into consideration. It can be accessed by pressing Alt+N.

Local time. Whenever a country in the box is selected, this parameter displays the local time of that country.

Base time. This converts the selected time into four bases: decimal (D), hexadecimal (H), binary (B) and octal (O). This function is nothing but a different way of representing the time in digital format.

Swatch Internet time. Swatch Internet time (or beat time) is a decimal time concept introduced in 1998 and marketed by the Swatch Corporation as an alternative decimal measure of time. One of the goals was to simplify the way people in different time zones communicate about time, by eliminating time zones altogether.



Instead of hours and minutes, the mean solar day is divided into 1000 parts called 'beats.' Each .beat lasts one minute and 26.4 seconds. Time is indicated as a 3-digit number, out of 1000, after midnight. So, @248 would indicate 248 .beats after midnight, representing 248/1000 of a day, i.e., just over 5 hours and 57 minutes. For clarity, refer the

equivalent parameters given below:

- 1 day=1000 .beats
- 1 hour=41.666 .beats
- 1 min=0.6944 .beats
- 1 sec=0.01157 .beats

There are no time zones in Internet time; instead, the new time scale of Biel mean time (BMT) is used, based on Swatch's headquarters in Biel, Switzerland and equivalent to Central European time, West Africa time, and UTC+1. Unlike civil time (the official time in a local region, e.g. Indian Standard Time), in most European countries, Internet time does not observe daylight saving time (daylight saving time is the practice of advancing clocks just after winter season so that evenings have more apparent daylight. Typically clocks are adjusted forward one